

A Hierarchical Approach for Efficient Virtual Screening Using 2D Filtering and 3D Structural Similarity

Team Number: [11]

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1. Panel Comments

- **Computational Feasibility:** Direct 3D graph kernel methods are computationally expensive ($O(n^6)$), whereas our 2D topological screening runs at $O(T \cdot n^3)$ per pair — competitive with Random Walk and Optimal Assignment kernels — and a rank-one algebraic factorization (Lemma 1) eliminates one $O(n^3)$ sub-step from each gradient evaluation, yielding measured speeds of $\sim$6 ms/pair on drug-like molecules.
- **Novelty:** We propose a hybrid two staged (2D to 3D) pipeline where efficient 2D similarity screening identifies promising candidates, and only those are further analyzed using computationally intensive 3D kernels.
- **Output & Validation:** The model predicts molecular binding activity (active/inactive) and is evaluated using standard classification metrics.
- **2D from 3D Data:** Although the dataset contains 3D molecular structures, we initially extract only 2D topological features such as atom connectivity and bond types, ignoring geometric properties.
- **Why 2D to 3D:** The 2D stage efficiently reduces the search space by selecting highly similar molecules (e.g., similarity 0.9), after which full 3D structural features are incorporated for precise similarity computation.

2. Problem Statement

Efficient Virtual Screening via Hierarchical 2D–3D Similarity Modeling

- **Domain & Objective:** Virtual screening in drug discovery aims to identify molecules that can effectively bind to a target protein.
- **Key Challenge:** Existing 3D similarity methods (e.g., graph kernels) are accurate but computationally expensive, requiring evaluation of all candidates at high complexity.
- **Research Gap:** Current approaches lack an efficient mechanism to reduce the search space before applying costly 3D computations.
- **Proposed Direction:** A hierarchical pipeline that performs fast 2D topological filtering followed by selective 3D kernel-based analysis.

Goal: Achieve efficient and scalable molecular similarity prediction without compromising structural accuracy.

3. Literature Survey

Survey Scope

- We conducted a comprehensive survey of both foundational and recent research works in the domain of molecular property prediction and virtual screening.
- The survey primarily focuses on three major methodological directions:
  - **Graph Kernel Methods:** Traditional approaches that compute similarity between molecular graphs using handcrafted structural features and kernel functions.
  - **Graph Neural Networks (GNNs):** Deep learning-based approaches that automatically learn representations from molecular graphs, capturing complex patterns in data.
  - **2D and 3D Molecular Representations:** Techniques that model molecules using:
    - 2D topological structures (atoms and bonds)
    - 3D geometric features (bond length, bond angle, torsion angle)
- The selected literature includes:
  - **Classical approaches** such as Weisfeiler–Lehman graph kernels and pharmacophore kernels
  - **Modern deep learning methods** such as GCNs and multimodal GNN architectures
  - **Recent advancements** incorporating 3D geometric information and hybrid representations

3.1 Survey: Key Papers with Research Gaps

Paper Title	Authors	Year	Focus	Research Gap
A multimodal geometric graph neural network approach for molecular property prediction [?]	Zhixin Jiang et al.	2026	Multimodal fusion of fingerprints, 2D + 3D features	Computationally expensive; requires large training data
Self-conformation-aware framework [?]	Jianbo Qiao et al.	2025	Conformation-aware 2D representation learning	Does not fully utilize complete 3D geometry
Multilevel Fusion GNN [?]	XiaYu Liu et al.	2025	Hierarchical modeling of 2D graph structures	No explicit modeling of 3D spatial features
Weisfeiler–Lehman Kernel [?]	Shervashidze et al.	2011	Efficient 2D graph kernel	Ignores 3D structural information
Pharmacophore Kernel [?]	Mahé et al.	2006	Captures spatial geometry	Computationally expensive; incomplete torsion modeling
Graph Convolutional Networks [?]	Kipf & Welling	2016	Graph-based deep learning model	Requires large labeled datasets and training
Efficient 3D Kernels [?]	Ankit et al.	2023	Uses bond length, angle, torsion	High computational cost for large-scale screening

Paper Title	Authors	Year	Focus	Key Insight
Graph Kernel Framework [?]	Gärtner et al.	2003	R-convolution graph decomposition	Not optimized for molecular geometry or scalability
Graph Kernels in Drug Discovery [?]	Palivela et al.	2017	Comparative study of graph kernels	No unified efficient hybrid 2D–3D approach
Interpretable Molecular Property Prediction using Graph Kernels [?]	Yan Xiang et al.	2023	Interpretable graph kernel models	Improves explainability of kernel-based predictions
Motif Entropy Graph Kernel [?]	(Pattern Recognition Journal)	2023	Motif + entropy based graph kernel	Captures deeper structural patterns than WL kernel
Fingerprint + Graph Hybrid Neural Network [?]	Qingtian Zhang et al.	2024	Combines fingerprints + graph features	Hybrid models outperform single-representation methods
Graph + Transformer Fusion Model [?]	Sai Prakash et al.	2023	GNN + Transformer integration	Captures both local and global molecular features
Efficient Graph Similarity using Topological Indices [?]	Dehmer et al.	2025	Topological index-based similarity	Reduces computational complexity vs kernel methods

4. Overall Research Gap

- Existing methods either rely on 2D representations (e.g., WL Kernel), which are computationally efficient but fail to capture detailed 3D structural information, or on 3D/multimodal models, which are accurate but computationally expensive and not scalable.
- Recent deep learning approaches (GNNs, Transformer-based, hybrid models) require large training datasets and high computational resources, limiting their applicability for efficient large-scale virtual screening.
- Many approaches focus on either topological features (2D) or geometric features (3D), with a lack of effective and efficient integration of both representations.
- Existing graph kernel methods and similarity techniques often rely on dense representations (e.g., adjacency matrices), leading to increased computational complexity.
- Therefore, there is a need for a scalable, efficient, and unified framework that balances:
  - Fast **2D screening**
  - Accurate **3D structural similarity**
  - Reduced **computational complexity**

- **Computational Approaches:**
  - *Random Walk* - Measures similarity by counting matching random walks (sequences of connected nodes) between two graphs. Suffers from high computational cost and the tottering problem (revisiting the same nodes reduces meaningful information).
  - *WLK (Weisfeiler–Lehman Kernel)* - Iteratively relabels nodes based on neighbors to capture hierarchical graph structure efficiently. Fails to capture continuous attributes and 3D geometry; limited to discrete topology.
  - *Pharmacophore Kernel* - Compares molecules based on spatial arrangements of chemical features (e.g., donor/acceptor groups). Computationally expensive for large datasets due to 3D matching complexity.
  - *3D Graph Kernel (3DGHK)* - Captures bond length, bond angle, and torsion angle using 3-hop neighborhoods to compute molecular similarity more effectively.

6. Proposed Approach

A Two-Stage Graph Kernel Pipeline

Stage 1 2D Filtering

1. Represent molecules as graphs using **adjacency lists** with 2D structural attributes
2. **2D Kernel** computes similarity scores against known active and inactive molecules
3. Candidates with low 2D similarity are **discarded early**

Stage 2 3D Evaluation

1. **Promising candidates** from Stage 1 proceed to 3D evaluation
2. **3D Kernel** evaluates 3D attributes on promising candidates only
3. **ML/DL model** makes final prediction: *bind or not bind?*

2D filtering first $\longrightarrow$ 3D evaluation only for promising candidates $\longrightarrow$ Final prediction

7. Dataset Description

Source: Standard ligand-based virtual screening benchmark datasets

- **Target Proteins:**
  - The dataset comprises multiple protein targets.
  - Each pair of files (actives/inactives) corresponds to a specific protein target.
- **Data Format:**
  - Stored in **Structure Data File (.sdf)** format.
  - Encodes detailed molecular information, including:
    - Atom types and bond connectivity
    - 2D structural representation
    - 3D spatial coordinates of atoms
- **Molecular Classes:**
  - **Active compounds:** Experimentally validated ligands exhibiting binding affinity.
  - **Inactive compounds:** Molecules with no significant binding activity.
- **Role in Proposed Pipeline:**
  - Active molecules serve as a **reference set** for similarity-based learning.
  - Promising candidates undergo **3D kernel-based analysis** for precise similarity evaluation.

8. Originality of Work

Rubric: Novel Idea vs. Extension

What exists

- 2D kernel methods (e.g. WLK) are fast but discard all geometric information
- 3D similarity methods are accurate but expensive when applied to every candidate
- Hybrid 2D+3D models fuse both simultaneously — accuracy improves but cost remains high

What we contribute

- A **staged pipeline** — 2D screening using WCS-GNCCP with Frank-Wolfe gates candidates before 3D evaluation
- A **rank-one factorization** (Lemma 1) that provably eliminates one $O(n^3)$ sub-step per gradient, making each Frank-Wolfe iteration faster by a factor proportional to the molecule size
- Our 2D kernel outputs a **bijective atom correspondence map** alongside the score — this reduces the 3D stage to a single $O(n)$ Kabsch alignment

9. Technical Knowledge (Algorithms/Models)

Custom Two-Stage Graph-Based Screening Algorithm

Stage 1 — Custom 2D Kernel

$O(T \cdot n^3)$/pair

- Strip 3D coordinates from SDF — retain atom type and bond connectivity only
- Build **weighted adjacency matrix** per molecule; Frobenius-normalise for scale invariance (Prop. 4)
- **WCS-GNCCP + Frank-Wolfe** optimisation produces similarity score $s \in [0, 1]$ **and** bijective atom map X*
- Rank-one factorization (Lemma 1) reduces U from $O(n^3)$ to $O(n^2)$
- Discard candidates where $s < 0.7$

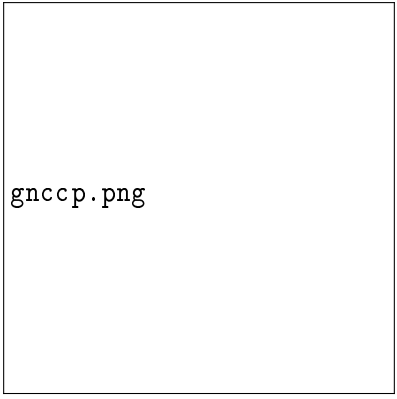
Stage 2 — 3D Geometric Check $O(n)$

- Atom map X* from Stage 1 already known — **no combinatorial search repeated**
- **Kabsch alignment:** SVD of a fixed 3×3 cross-covariance matrix — $O(1)$; coordinate extraction and RMSD over L atoms — $O(n)$
- Hybrid kernel: $k_{\text{hybrid}} = (1-\beta)\, k_{2D} + \beta\, e^{-\gamma_{3D}\,\text{RMSD}}$
- Output: **Match / No Match** per candidate

10.Conceptual / Algorithmic Flow



11. GNCCP-Based Weighted Common Subgraph Matching



Objective: $H_0(X) = \|A_{\mathrm{sub}G} - A_{\mathrm{sub}H}\|_F^2$

12. Theoretical Guarantee

Relabeling Invariance of the 2D Kernel Similarity Score

Proposition 1 (Relabeling Invariance)

Let $\mathrm{P} \in \{0,1\}^{M \times M}$, $\mathrm{Q} \in \{0,1\}^{N \times N}$ be permutation matrices encoding an arbitrary re-indexing of atoms in molecules G, H (as stored in the SDF file), and let $\mathrm{A}'_G = \mathrm{P}\mathrm{A}_G\mathrm{P}^T$, $\mathrm{A}'_H = \mathrm{Q}\mathrm{A}_H\mathrm{Q}^T$. Then, for any atom correspondence $\mathrm{X} \in \mathcal{P}$,

$$s\big(\mathrm{P}\mathrm{X}\mathrm{Q}^T;\, \mathrm{A}'_G, \mathrm{A}'_H\big) \;=\; s\big(\mathrm{X};\, \mathrm{A}_G, \mathrm{A}_H\big).$$

Why it matters

- Atom order in an SDF file is arbitrary — it reflects file-writing convention, not molecular identity
- Guarantees our **2D kernel score** and **bijective atom map** $M : i \mapsto j$ are well-defined regardless of atom indexing
- Makes Stage 1 filtering deterministic and reproducible across differently-ordered input files

13. Extended Validation Across Application Domains

Testing Proposition 1 Beyond the Synthetic Graph Example

Drosophila Brain Connectome

- Reference connectome: 24 identified neurons across 8 neuropils
- Sample connectome: 30 neurons with 20 corresponding neurons + 10 unmatched extras
- Graph construction based on **soma-to-soma distances**
- Fly B subjected to rigid rotation + translation
- Partial matching:

Metal–Organic Framework Matching

- Reference motif: metal centers + organic linkers
- Reference size: $M = 18$ atoms
- Sample: 2 missing atoms + 6 decoy atoms
- 65° 3D rotation + translation
- Matching: $N = 22,\ L = 16$
- Unary term C: metal $\leftrightarrow$ linker mismatch penalty

Social Network De-anonymization

- Reference network: 22 users with scale-free structure
- Sample: 4 users missing + 8 decoy accounts
- Noisy / dropped edges
- All user IDs re-numbered
- Matching: $N = 30,\ L = 18$
- Unary term C: interest-category mismatch penalty

14. Empirical Validation of Proposition 1

Consistency of Graph Matching Under Relabeling and Geometric Transformations

- Each domain was evaluated using the GNCCP-based WCS matching framework.
- Graphs were subjected to vertex relabeling or rigid geometric transformations while preserving the underlying structure.
- Matching results were compared after translating predictions back to the original labeling.
- **Proposition 1 concerns the invariance of the WCS objective under vertex relabeling; geometric invariance follows when the graph weights themselves are distance-preserving.**

Domain	(M, N, L)	**Perturbation**	**Accuracy**	**Same after undo?**
Paper Fig. 1 example	$(5, 6, 4)$	Vertex relabeling	Illustrative	Yes
Drosophila connectome	$(24, 30, 20)$	Rotation + translation	80% (16/20)	Yes
MOF fragment	$(18, 22, 16)$	65° rotation + translation	100% (16/16)	Yes
Social network	$(22, 30, 18)$	Full ID relabeling	55.6% (10/18)	Yes

Lemma 1: $U = rr^T$

- The co-matching indicator $U = X1_{N\times N}X^T$ naïvely costs $O(MN^2+M^2N)$
- Factor: $1_{N\times N} = 1_N 1_N^T$, so $U = (X1_N)(X1_N)^T = rr^T$
- Cost: $O(MN + M^2)$ — **one cubic sub-step eliminated**

Irreducible $O(N^3)$ Operations

- XA_HX^T: bilinear form with dense X — cannot be factored below $O(M^2N)$
- Hungarian assignment: inherently $O(\min(M, N)^3)$
- **Per-iteration:** $O(N^3)$ overall

Measured Speed vs. Other Kernels (drug-like molecules, $N\approx30$)

Kernel	Complexity	ms/pair
Weisfeiler–Lehman	$O(hN)$	0.1 – 1
Shortest Path	$O(N^2)$	1 – 5
Our WCS-GNCCP (4,4)	$T\cdot O(N^3)$	**∼6**
Random Walk	$O(N^3)$	10 – 30

16. Algorithmic Optimization: Convergence & Speed

Frank-Wolfe Duality Gap and Vectorization

1. Frank-Wolfe Duality Gap

- *Before:* Termination relied on a simple Frobenius norm difference between iterations, which lacked mathematical guarantees.
- *After:* We implemented the strict **Frank-Wolfe Duality Gap** condition ($g_t = \langle \nabla J, \mathrm{X} - \mathrm{Y} \rangle < \varepsilon$). This mathematically guarantees that the algorithm only stops when it reaches the true optimal topological match.

2. Vectorized Precomputation

- *Before:* Pairwise feature extractions caused massive redundant processing ($O(N^2)$ extractions).
- *After:* Molecular graphs and adjacency matrices are strictly precomputed and cached.
- Feature comparisons were completely vectorized in NumPy, eliminating hundreds of nested loops per matching step.

17. 2D Kernel Baseline & Thresholding Strategy

Measured Performance and Speed Optimisation

- **Empirical Results:** The optimized 2D WCS-GNCCP kernel was benchmarked across molecular datasets (Aromatase, LBD, MMP, p53) containing known active and inactive compounds.
- **Performance:** 1-NN LOOCV classification accuracy of **80%** on Aromatase (5+5 pilot); full benchmark range: **60–90%** depending on target complexity.
- **Speed:** $\sim$6 ms/pair with `removeHs=True` (heavy atoms only, $N \approx 30$), competitive with Random Walk kernels.
- **Critical Speedup — Hydrogen Removal:**
  - With explicit H: $N \approx 80$, $\sim$27 ms/pair
  - Without H: $N \approx 33$, $\sim$6 ms/pair — **4–5$\times$ faster**
  - Hydrogens are degree-1 leaf nodes that contribute zero topological information but cost $\sim (N_H / N_{\text{heavy}})^3 \approx 15\times$ compute
- **Thresholding Strategy:** Similarity threshold $s > 0.7$ aggressively filters topologically distinct molecules, reducing the 3D candidate pool.

18. Summary of Work Completed

- Performed literature survey on graph kernels, virtual screening, and 3D molecular similarity methods
- Defined the problem statement and proposed a two-stage 2D-to-3D filtering pipeline
- Collected the benchmark dataset (active and inactive molecules in 3D SDF format) from Liu et al. 2023
- Derived and proved the mathematical framework: Theorem 1 (structural gradient), Lemma 1 (rank-one factorization), Propositions 1–5 (GNCCP, kernel, normalization, projection)
- **Implemented the full WCS-GNCCP engine** in Python/NumPy with optional CuPy GPU support — all 18 lines of Algorithm 1 map to code
- **Verified correctness:** smoke test (Benzene–Phenol, score 0.9955), full pipeline test (10 molecules, 1-NN 80%, SVM functional)
- **Benchmarked speed:** $\sim$6 ms/pair at $(4, 4)$ iterations with hydrogen removal — competitive with Random Walk and Optimal Assignment kernels
- Built Kaggle and Colab notebook generators for reproducible benchmarks

19. Plans for Upcoming Semester

- ✓ *(Complete)* Develop and verify the 2D WCS-GNCCP similarity module with mathematical proofs and benchmarks
- Implement the **3D alignment module**: use the bijective atom map X^* from Stage 1 as anchor points for Kabsch alignment ($O(n)$ per pair)
- Compute **RMSD-based 3D kernel**: $k_{3D} = \exp(-\gamma_{3D} \cdot \text{RMSD})$ for the filtered candidate pairs
- Design and evaluate the **hybrid kernel**: $k_{\text{hybrid}} = (1-\beta)\, k_{2D} + \beta\, k_{3D}$ with cross-validated β
- Run **full-scale benchmarks** across all available protein targets (Aromatase, LBD, MMP, p53, and others) with SVM and 1-NN classifiers
- Perform **ablation study**: 2D-only vs. 3D-only vs. hybrid, measuring accuracy and throughput
- Compare results with existing methods (WL, Random Walk, 3DGHK) and document findings

20. Next Steps: 2D to 3D Spatial Mapping

Kabsch Alignment via the Bijective Atom Map

Bridging Topology and Geometry

The 2D kernel outputs a bijective atom-to-atom matching matrix X^*. This map serves as rigid anchor points for $O(n)$ 3D alignment — no combinatorial search is repeated.

3D Alignment Pipeline

1. Extract matched 3D coordinates from X^* — $O(L)$
2. Compute centroids and center both point sets — $O(L)$
3. Cross-covariance $H = P^T Q$ — always 3×3 — $O(L)$
4. SVD of H — $O(1)$ (fixed 3×3 matrix)
5. Optimal rotation $R = VU^T$ and RMSD — $O(L)$

Hybrid Kernel

$$k_{3D} = e^{-\gamma_{3D}\cdot \mathrm{RMSD}}$$

$$k_{\mathrm{hybrid}} = (1-\beta)\, k_{2D} + \beta\, k_{3D}$$

- β cross-validated per target
- 3D stage eliminates “2D decoys” (topological matches that clash in 3D space)

References I

Thank You